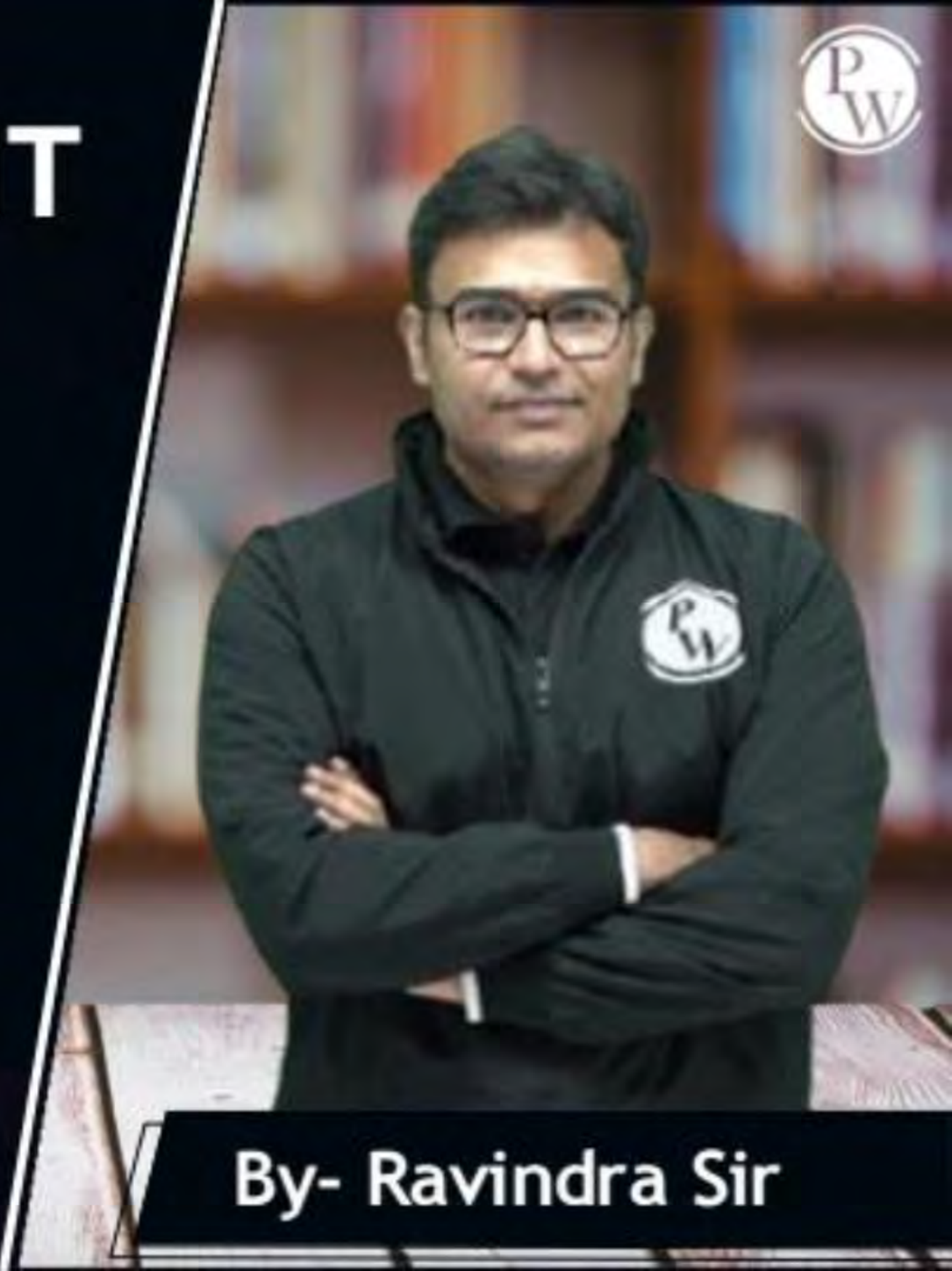
Computer Science & IT

COMPILER DESIGN
(CD)

SDT

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Parsec

Topics to be Covered



- Topic
- Topic
- Topic

SDT

Join Telegram

@RBRSIRPW





Inspiring Stories : Peer Ali Khan

Peer Ali Khan Background: Born around 1812 in Azamgarh, Uttar Pradesh. Worked as a bookbinder in Patna.
Struggle: Saw British exploiting and humiliating Indians daily. Became part of a secret rebel network before 1857 revolt.
Education: Learned to read and write Urdu and Persian from local teachers.
Achievement: In 1857, helped plan the uprising in Patna — stored weapons, spread coded messages hidden inside books, and coordinated rebels. When the revolt broke out, he was caught with letters exposing the entire plan. Refused to name a single comrade.
Impact: Publicly hanged in July 1857 while calmly smiling at the crowd. His fearlessness under death inspired locals for decades — but his name was buried in forgotten pages.



Inspiring Stories : Velu Nachiyar

Velu Nachiyar Background: Born 1730 in Tamil Nadu, a princess trained in weapons and warfare.

Struggle: British killed her husband in battle and took over her kingdom.

Education: Trained in horse riding, sword fighting, archery, and military strategy since childhood.

Achievement: Built an army, made alliances, and planned an attack. Her secret weapon: her commander Kuyili soaked herself in oil, lit herself on fire, and jumped into the British ammunition store, blowing it up. Velu Nachiyar won back her kingdom — 80 years before 1857.

Impact: India's first woman ruler to fight and defeat the British. A living example of patience, planning, and revenge served hot.



Inspiring Stories : Tirot Sing

Tirot Sing Background: Born in the early 1800s, chief of the Khasi Hills in today's Meghalaya.

Struggle: British tried to grab his land by fooling him with a fake "friendship" treaty.

Education: No school — learned leadership, planning, and war skills from his tribe's elders.

Achievement: When he found out the treaty was a trick, he declared war. For four years, he attacked the British like lightning — sudden, fast, from nowhere — in forests and hills. Before they could fight back, he would vanish. He drained their money, soldiers, and courage.

Impact: Caught in 1833 and died in a British prison. In Meghalaya, he is still remembered as the chief who refused to bow, showing that brains and speed can beat an empire.



Inspiring Stories : U Kiang Nangbah

U Kiang Nangbah Background: Born in the 1830s in the Jaintia Hills of Meghalaya.

Struggle: Watched the British loot his land and insult his people's culture and traditions.

Education: No formal school — learned wisdom, strategy, and courage from his elders.

Achievement: In 1862, united his people and rose up against the British. For two years, struck them like a thunderclap — sudden attacks, disappearing before they could hit back. Finally caught and sentenced to hang. On the gallows, he told the crowd: "If my face turns east when I die, my country will be free soon."

Impact: His last words became a prophecy and a legend. Even today in the Northeast, his name burns as a symbol of pride, courage, and never bending to an enemy.

Syntax directed translations :-

SDT is an extension of CFGS. This helps the Compiler design to translate language constructs directly by attaching Semantic actions of Subroutines.

Grammar + Semantic rules = SDT

With SDT, we can combine many tasks with parsing

Along with each production of grammar we attach Semantic actions

Some examples of tasks performed by SDT's are

1) To store/retrieve type information in Symbol table
2) To perform consistency checks like type checking, parameter checking, etc
3) To issue error messages
4) To build Syntax trees
5) To generate intermediate code or target code

→ In SDT, we parse, create parse trees and traverse the parse tree as needed to evaluate Semantic actions at parse tree nodes

SDTs are called attribute grammars

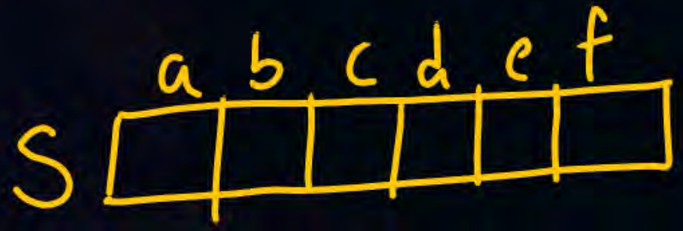
Where

1) Each grammar is assigned certain attributes
2) Each production is augmented with semantic rules which are used to define attribute values

Attributes for grammar Symbols :-

SDT is a generalization of a CFG in which each grammar Symbol has an associated set of attributes.

If we consider parse tree node as a record for holding information, then an attribute is the name for each field in the record



A grammar Symbol can have 'n' attributes. The attributes for the grammar can be a type, value, address, pointer of a string. The attributes that can be associated with the grammar Symbol can be classified into two types.

Attribute types :-

1. Synthesized attribute :- The value of a synthesized attribute at a node is computed from the values of attributes at the children of that node in the parse tree

$$A \rightarrow XYZ$$

Let us assume there is an attribute 'S' with each grammar symbol.

Then if is evaluated as

$$A.S = f(X.S, Y.S, Z.S)$$

The 'S' is called Synthesized attribute

2) Inherited attribute :- The value of an inherited attribute is computed from the values of attributes at the siblings and parent of that node

Ex:- $A \rightarrow XYZ$

Assume there is an attribute 'i' with each grammar symbol then it is evaluated as $Y.i = f(X.i, Z.i, A.i)$ then i is called the inherited attributes

→ An attribute will be either synthesized or inherited.

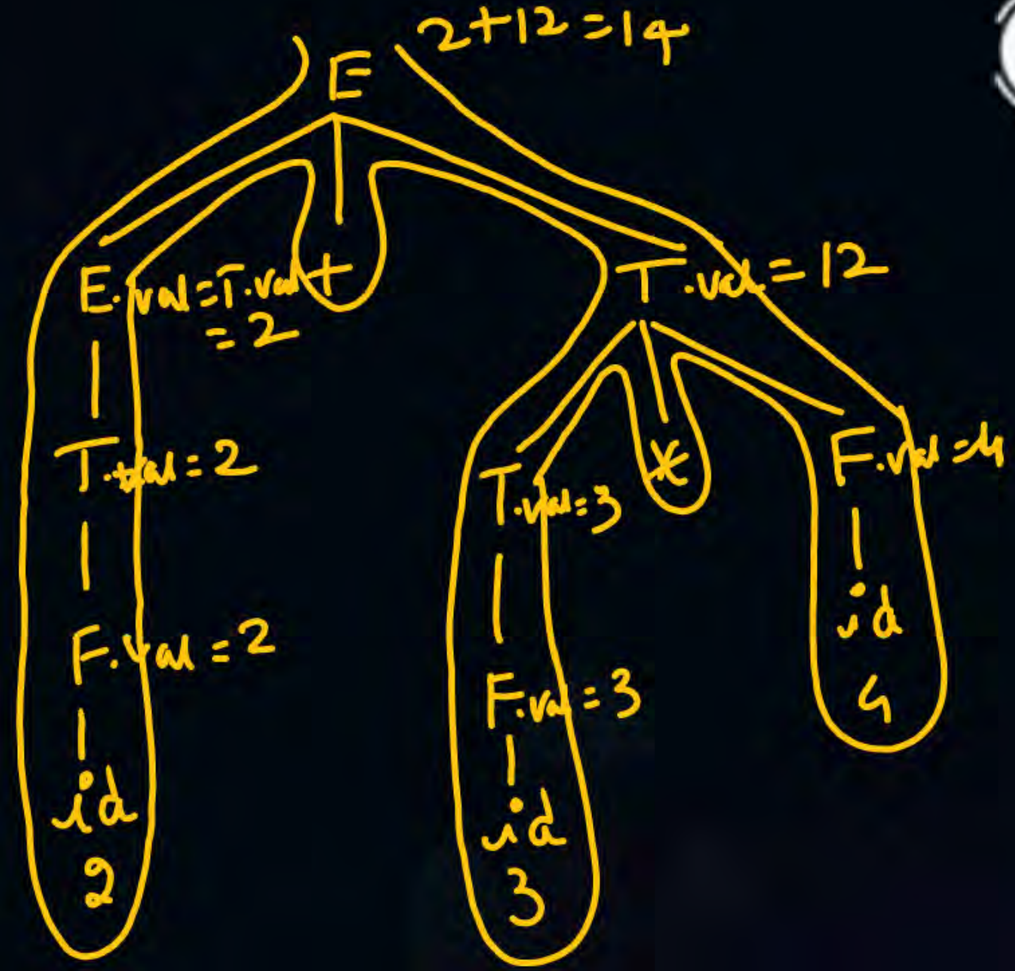
→ It cannot be both.

→ SDTs are generally used with BUP.

→ We assume that whenever a BUP reduces by a production, the corresponding action is automatically carried out

$E \rightarrow E_1 + T$	$\{E.val = E_1.val + T.val\}$
$/T$	$\{E.val = T.val\}$
$T \rightarrow T_1 * F$	$\{T.val = T_1.val * F.val\}$
$/F$	$\{T.val = F.val\}$
$F \rightarrow id$	$\{F.val = id.lval\}$

$2 + 3 * 4$



SDT to convert infix to postfix:-

$E \rightarrow E+T$ { Printf "+" } ①

$/T$ { } ②

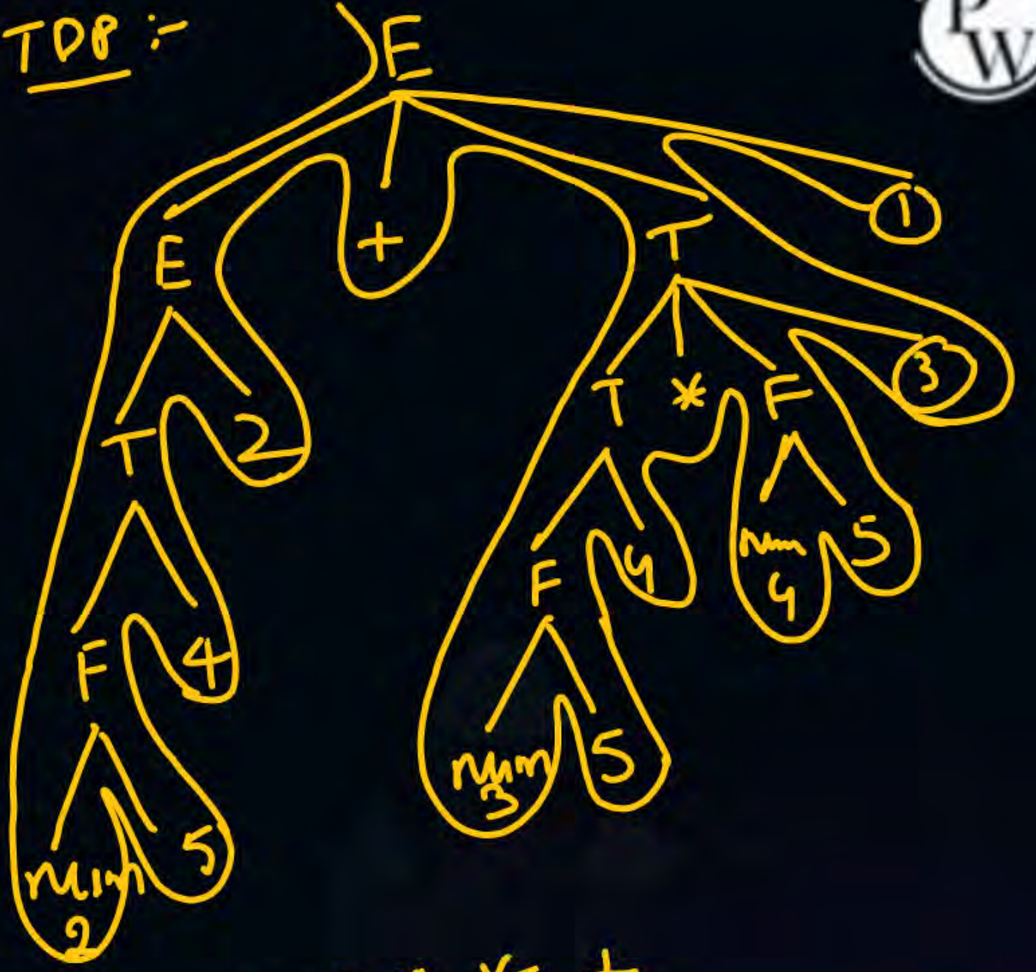
$T \rightarrow T*F$ { Printf '*' } ③

$/F$ { } ④

$F \rightarrow num$ { Print(num.lval), } ⑤

2 + 3 * 4

TDB :-



2 3 4 * +

2+3*4

How BUP :-



2 3 4 * +

$S \rightarrow x\,x\,w$ { Print '1' }

$\quad / y$ { Print '2' }

$w \rightarrow S\,z$ { Print '3' }

$x\,x\,x\,x\,y\,z\,z$



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$E \to E_1 * T$ { E.val = E_1.val * T.val }

/T { E.val = T.val }

$T \to F - T$ { T.val = F.val - T_1.val }

/F

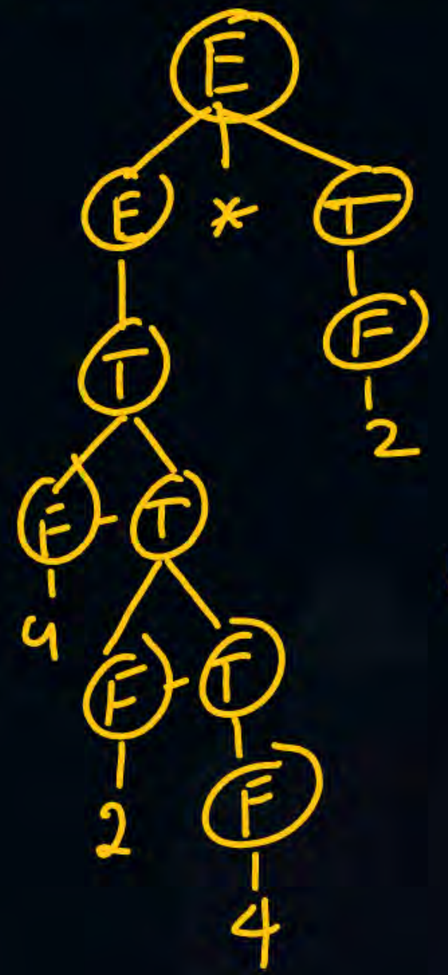
$F \to 2$ { F.val = 2 }

/4 { F.val = 4 }

$$w = \frac{(4-(2-4)*2}{4-(-2)*2}$$

$6 * 2 = 12$

Count the no of reductions



10 reductions

$E \rightarrow E_1 \# T$ $\{E.val = E_1.val * T.val\}$

$/T$ $\{E.val = T.val\}$

$T \rightarrow T \& F$ $\{T.val = T.val + F.val\}$

$/F$ $\{T.val = F.val\}$

$F \rightarrow num$ $\{F.val = num.lval\}$

$2 \# 3 \& 5 \# 6 \& 4$

$2 * (3 + 5) * (6 + 4)$

$(2 * 8) * 10$

$\underline{160}$

THANK - YOU